MeepleAI - Feature e Changelog

Descrizioni feature, changelog, PDF processing queue, agent chat RAG integration.

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File inclusi: 3

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features/agent-chat-rag-integration.md

Agent Chat with RAG Integration

Overview

Complete implementation of agent chat functionality with RAG-powered responses. Enables users to chat with AI agents that have access to game-specific knowledge base documents.

Status: ✅ Production Ready **Version**: 1.0 **Date**: 2026-02-17

Architecture

System Flow

(blocco di codice rimosso)

Data Model

(blocco di codice rimosso)

API Reference

GET /api/v1/agents/{id}/status

Purpose: Check if agent is ready for chat (KB validation).

Authentication: Required (session)

Response (200): *(blocco di codice rimosso)*

Response (Not Ready): *(blocco di codice rimosso)*

Error Codes:

- `404`: Agent not found

POST /api/v1/agents/{id}/chat

Purpose: Chat with agent using SSE streaming with RAG context.

Authentication: Required (session)

Request: *(blocco di codice rimosso)*

Response: SSE stream with events

Event Types:

Type	Code	Data	Description
StateUpdate	0	`{ message, chatThreadId }`	Status message
Citations	1	`[{ documentId, pageNumber, score }]`	RAG citations
Token	7	`{ token }`	Streamed text token
Complete	4	`{ totalTokens, chatThreadId, ... }`	Stream complete
Error	5	`{ errorMessage, errorCode }`	Error occurred

Error Codes:

- `AGENT_NOT_FOUND`: Agent ID invalid
- `AGENT_NOT_CONFIGURED`: No AgentConfiguration found
- `AGENT_NO_DOCUMENTS`: SelectedDocumentIds empty
- `EMBEDDING_FAILED`: Embedding service error
- `SEARCH_FAILED`: Qdrant search error
- `THREAD_NOT_FOUND`: ChatThreadId invalid

Example SSE Stream: *(blocco di codice rimosso)*

Frontend Integration

useAgentStatus Hook

Purpose: Check agent readiness before enabling chat.

Usage: *(blocco di codice rimosso)*

Return Type: *(blocco di codice rimosso)*

AgentInfoCard Component

Location: `apps/web/src/components/agent/AgentInfoCard.tsx`

Features:

- ✅ 3 tabs: Chat, History, Knowledge Base
- ✅ Auto status check on mount
- ✅ Blocking UI if agent not ready
- ✅ Embedded ChatThreadView when ready
- ✅ Fullscreen mode toggle
- ✅ Auto-thread creation on first message

Props: *(blocco di codice rimosso)*

States: *(blocco di codice rimosso)*

Validation Rules

Agent Readiness Requirements

Agent is **ready for chat** when ALL conditions are met:

1. ✅ `Agent.IsActive == true`
2. ✅ `AgentConfiguration` exists with `IsCurrent == true`
3. ✅ `AgentConfiguration.SelectedDocumentIds.Count > 0`
4. ✅ Documents are in Qdrant (implicit - checked during search)

Error Handling Matrix

Condition	Backend Error	Frontend UI
Agent not found	404	"Agent not found" error page
Agent inactive	AGENT_NOT_FOUND	N/A (filtered from list)
No configuration	AGENT_NOT_CONFIGURED	"Agent non configurato" blocking UI
No documents	AGENT_NO_DOCUMENTS	"Agent non ha documenti nella KB" blocking UI
Embedding fails	EMBEDDING_FAILED	SSE error event → error banner
Search fails	SEARCH_FAILED	SSE error event → error banner

RAG Pipeline Details

Step-by-Step Process

1. Validation (SendAgentMessageCommandHandler:90-129): *(blocco di codice rimosso)*

2. Embedding Generation (SendAgentMessageCommandHandler:183-194): *(blocco di codice rimosso)*

3. Vector Search (SendAgentMessageCommandHandler:196-215): *(blocco di codice rimosso)*

4. Context Building (SendAgentMessageCommandHandler:217-227, 311-320): *(blocco di codice rimosso)*

5. LLM with Context (SendAgentMessageCommandHandler:248-264): *(blocco di codice rimosso)*

6. Citations Emission (SendAgentMessageCommandHandler:229-242): *(blocco di codice rimosso)*

Configuration

Agent Configuration Requirements

For **Player** or **Ledger** modes:

- **MUST** have `SelectedDocumentIds.Count >= 1` (enforced in AgentConfiguration.cs:154)

For **Chat** mode:

- **RECOMMENDED** to have documents for RAG responses
- Falls back to generic LLM if no documents (with user-visible warning)

Search Parameters

Configurable (currently hardcoded in SendAgentMessageCommandHandler:218): *(blocco di codice rimosso)*

Future Enhancement: Load from `AgentConfiguration.Strategy` parameters.

Testing

E2E Test Coverage

File: `apps/web/e2e/agent/agent-chat-page-rag.spec.ts`

Test Cases (7 scenarios):

1. ✅ Blocking UI when agent has no KB documents
2. ✅ Chat enabled when agent is ready
3. ✅ Thread creation and embedding
4. ✅ RAG-powered SSE streaming with citations
5. ✅ Fullscreen mode toggle functionality
6. ✅ SSE connection maintained during fullscreen
7. ✅ KB tab displays documents correctly

Error Scenarios (2 tests): 8. ✅ Agent not found (404 handling) 9. ✅ Status check failure (500 handling)

Responsive (1 test): 10. ✅ Mobile viewport adaptation

Run Tests: *(blocco di codice rimosso)*

Performance Considerations

RAG Pipeline Latency

Expected Timings:

- Embedding generation: ~100-200ms (local sentence-transformers)
- Vector search (Qdrant): ~50-150ms (depends on collection size)
- Context building: <10ms (in-memory)
- LLM streaming: 2-5s (depends on response length + model)

Total: ~2.5-5.5s for complete response

Optimization Opportunities

1. **Parallel Operations**: Embedding + Config loading can be parallelized
2. **Caching**: Cache AgentConfiguration per session
3. **Streaming Start**: Start LLM streaming while still searching (speculative execution)

Troubleshooting

Common Issues

"Agent non configurato":

- **Cause**: No AgentConfiguration with `IsCurrent = true`
- **Fix**: Admin → Configure agent via `/admin/ai-lab/agents/{id}/edit`

"Agent non ha documenti nella KB":

- **Cause**: `SelectedDocumentIds` is empty
- **Fix**: Admin → Add documents to agent KB

"Embedding generation failed":

- **Cause**: Embedding service down or network error
- **Fix**: Check `embedding-service` container health
- **Check**: `docker logs meepleai-embedding-service`

"Vector search failed":

- **Cause**: Qdrant collection doesn't exist or network error
- **Fix**: Check Qdrant health, verify game documents indexed
- **Check**: `docker logs meepleai-qdrant`

"No relevant context found":

- **Cause**: All chunks scored below minScore (0.6)
- **Behavior**: LLM responds with "No relevant context found in knowledge base"
- **Fix**: Check if documents are properly indexed, adjust minScore threshold

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Future Enhancements

Phase 2 Improvements

1. **Dynamic Parameters**:
   - Load `minScore`, `topK` from AgentConfiguration.Strategy
   - Per-agent tuning of retrieval parameters
2. **Hybrid Search**:

- Integrate `HybridSearchEngine` (BM25 + Vector + Reranking)
- Improve relevance with keyword matching

3. **Multi-Turn Context**:
   - Include previous messages in embedding
   - Maintain conversation context for follow-up questions
4. **Citation Click Handling**:
   - Navigate to PDF viewer at specific page
   - Highlight relevant chunk in document
5. **Response Quality Metrics**:
   - Track confidence scores
   - User feedback integration (thumbs up/down)

Code Refactoring (Task #6)

Extract RAG logic into reusable service: *(blocco di codice rimosso)*

Benefits:

- Testability (mock service in unit tests)
- Code reuse (multiple handlers can use same logic)
- Easier to add caching layer

Migration Guide

For Existing Agent Configurations

No migration needed - backward compatible:

- Agents without documents → Blocking UI prevents chat
- Agents with documents → Chat works immediately
- Existing ChatThreads → AgentId nullable, works without it

For Frontend Components

Breaking Changes: None

New Features Available:

- Use `useAgentStatus(agentId)` to check readiness

- Use `AgentInfoCard` with embedded chat
- Pass `agentId` in `createThread()` for agent-specific threads

Related Documentation

- **RAG System**: `docs/ai/rag-architecture.md`
- **Agent Configuration**: `docs/features/agent-configuration.md`
- **SSE Streaming**: `docs/api/sse-streaming.md`
- **Vector Search**: `docs/ai/vector-search.md`
- **Chat System**: `docs/features/chat-threads.md`

Changelog

v1.0 (2026-02-17) - Initial Implementation:

- ✅ RAG integration in SendAgentMessageCommandHandler
- ✅ Agent status validation endpoint
- ✅ Embedded chat in AgentInfoCard
- ✅ Fullscreen mode support
- ✅ E2E test coverage
- ✅ KB empty blocking UI

Files Modified: 11 (5 backend, 5 frontend, 1 test)

Lines of Code: ~600 LOC added/modified

Test Coverage: 10 E2E scenarios + existing unit tests

MeepleAI — Changelog Funzionalità (21 Gen – 21 Feb 2026)

> *Riepilogo di pagine, endpoint e funzionalità aggiunte o modificate nell'ultimo mese. Data generazione: 2026-02-21*

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1. Funzionalità per Area
2. Pagine Frontend — Inventario Completo
3. Endpoint Backend Aggiunti

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1. Funzionalità per Area

Chat & AI Agents

#	Funzionalità	Issue
1	**Custom Agents**: wizard di creazione agente, AgentCreationSheet, GameSelector/ModelSelector collegati a API reale	#4804
2	**Chat grouping**: sessioni raggruppate per agente con limiti tier-based	#4913
3	**Tab Debug chat**: visibilità pipeline RAG direttamente in chat	#4916
4	**PDF Citations in chat**: citazioni PDF nelle risposte AI	#4912
5	**Agent quota**: enforce creazione agenti per tier utente	#4944
6	**Chat welcome message** + proxy SSE OpenRouter	#4780
7	**Auto-link PDF**: collegamento automatico PDF indicizzati al game agent	#4941

Admin Dashboard

#	Funzionalità	Issue
1	**Navigation redesign**: Universal Navbar + Context-Sensitive Sidebar	#4936
2	**Debug Chat admin**: real-time RAG pipeline tracing	—
3	**Knowledge Base overview**: pagina panoramica + coda di elaborazione	#4892
4	**KB settings**: dashboard read-only + cache clear	#4888
5	**Processing metrics**: dashboard con P99 bug fix	#4880
6	**RAG pipeline health**: overview live	#4879
7	**Embedding Service**: dashboard + monitoring	#4878
8	**Qdrant operations**: operazioni avanzate + delete	#4877
9	**Upload & Process**: pagina wired con dati reali	#4876
10	**Documents Library**: gestione PDF admin	#4795
11	**Vector Collections**: dati reali Qdrant	#4793
12	**Chat history tier limits**: config admin	#4918
13	**Shared Games migration**: integrazione nel nuovo dashboard	—
14	**Card Stack panel** + table view con dati ruolo reali	#4814

Frontend Pages (nuove)

Pagina	Descrizione	Issue
`/profile`	Landing con tabs (achievements, stats, history)	#4893
`/games/[id]/strategies`	Strategie per il gioco	#4889
`/games/[id]/reviews`	Recensioni community	#4889
`/players/[id]/*`	Sub-route: achievements, games, sessions, stats	#4890
`/sessions/[id]/players`	Gestione giocatori sessione	#4891
`/sessions/[id]/notes`	Note di sessione	#4891
`/games/[id]/add` → AddGameSheet	Drawer wizard 3-step per aggiungere giochi	#4818–#4821
`/dashboard`	Redesign dashboard user-facing	#4936

Session Live Game (Player Journey)

#	Funzionalità	Issue
1	**LiveGameSession aggregate** + domain model	#4747
2	**SSE streaming infrastructure** + state broadcasting	#4764
3	**Session Join** via codice + ruoli attivi	#4766
4	**Player actions** + host validation + conflict resolution	#4765
5	**SSE Client** + Player/Spectator mode UI	#4767
6	**ExtraMeepleCard** (4 tab: Info, Media, AI, Actions)	#4757, #4762, #4763
7	**SessionSnapshot** history slider + time travel mode	#4755, #4758
8	**Turn Phases** + event-triggered snapshots	#4761
9	**SessionMedia** entity + RAG agent integration	#4760
10	**GameToolkit** bounded context + ToolState	#4753, #4754

Knowledge Base / RAG

#	Funzionalità	Issue
1	**KB Processing Status API** per game	#4943
2	**Notifica utente** quando PDF indicizzato	#4942
3	**SharedGameId** su VectorDocument per contenuti admin	#4921
4	**Admin PDF upload** (bypass user library, 500MB) + AgentDefinition.KbCardIds	#4922–#4925
5	**Orchestrated agent creation** con auto-setup flow	#4800
6	**Agent slots endpoint** con quota tier-based	#4799

MeepleCard Design System

#	Funzionalità	Issue
1	Migration: PrivateGameCard, SharedLibrary, RecentLibrary, BggCard, ShareRequestCard, VectorCollectionCard	#4857–#4869
2	**MeepleCard Session Front/Back**	#4751, #4752
3	**CardAgentAction footer**	#4777
4	**Responsive flip trigger**: touch button / desktop click	#4841
5	**3-state button visibility** system	#4899
6	**Uniform** su dashboard, /games e admin	#4909

2. Pagine Frontend — Inventario Completo

Pubbliche (nessuna autenticazione)

URL	Descrizione
`/`	Landing page marketing
`/about`	Informazioni sul prodotto
`/blog`	Blog post
`/contact`	Modulo contatto
`/contributions`	Contributi community
`/cookies`	Cookie policy
`/privacy`	Privacy policy
`/terms`	Termini di servizio
`/faq`	Domande frequenti
`/how-it-works`	Tutorial / funzionamento
`/discover`	Scoperta giochi consigliati
`/gallery`	Galleria visuale
`/offline`	Pagina offline fallback
`/games`	Catalogo giochi
`/games/catalog`	Vista alternativa catalogo
`/games/[id]`	Dettaglio gioco
`/games/add`	Aggiungi gioco al catalogo
`/shared-games/[id]`	Dettaglio gioco condiviso
`/giochi/[id]`	Dettaglio gioco (IT)
`/board-game-ai`	Info su AI per board game
`/board-game-ai/games`	Giochi con feature AI
`/sessions`	Lista sessioni pubbliche
`/sessions/[id]`	Dettaglio sessione

`/sessions/[id]/state`	Stato sessione (viewer)
`/sessions/join/[token]`	Join sessione via token
`/sessions/history`	Storico sessioni pubbliche
`/settings`	Impostazioni guest
`/settings/usage`	Quota e utilizzo
`/library/shared/[token]`	Libreria condivisa via token
`/preview/chat-card`	Preview componente chat card
`/preview/collection-dashboard`	Preview dashboard collection
`/design/cards`	Demo MeepleCard
`/demo/entity-list-view`	Demo EntityListView
`/demo/entity-list-complete`	Demo lista entità completa

Autenticazione (solo utenti non loggati)

URL	Descrizione
`/login`	Login
`/register`	Registrazione
`/welcome`	Benvenuto post-registrazione
`/reset-password`	Reset password
`/verify-email`	Verifica email
`/verification-pending`	In attesa verifica
`/verification-success`	Email verificata
`/oauth-callback`	Callback OAuth
`/auth/callback`	Callback auth generico

Utente Autenticato

Dashboard & Hub

URL	Descrizione
`/dashboard`	Gaming hub principale
`/dashboard/budget`	Overview budget/spesa

Chat

URL	Descrizione
`/chat`	Lista chat (raggruppate per agente)
`/chat/new`	Nuova conversazione
`/chat/[threadId]`	Thread chat attivo
`/chat/agents/create`	Crea agente chat

Agenti

URL	Descrizione
`/agents`	Lista agenti disponibili
`/agents/[id]`	Dettaglio agente
`/agent/slots`	Slot agenti disponibili

Libreria & Giochi

URL	Descrizione
`/library`	Libreria giochi
`/library/games/[gameId]`	Gioco in libreria
`/library/games/[gameId]/agent`	Agente AI per gioco
`/library/games/[gameId]/toolkit`	Toolkit per gioco
`/library/games/[gameId]/toolkit/[sessionId]`	Sessione toolkit attiva
`/library/private`	Collezione privata
`/library/private/add`	Aggiungi gioco privato
`/library/wishlist`	Lista desideri
`/library/proposals`	Proposte giochi
`/library/propose`	Proponi gioco

Giochi (Autenticato)

URL	Descrizione
`/games/[id]/faqs`	FAQ del gioco
`/games/[id]/reviews`	Recensioni community *(nuovo)*
`/games/[id]/rules`	Regole dettagliate
`/games/[id]/sessions`	Sessioni per questo gioco
`/games/[id]/strategies`	Strategie *(nuovo)*

Sessioni

URL	Descrizione
`/sessions/[id]/notes`	Note sessione *(nuovo)*
`/sessions/[id]/players`	Giocatori sessione *(nuovo)*

Play Records

URL	Descrizione
`/play-records`	Partite registrate
`/play-records/new`	Nuova partita
`/play-records/[id]`	Dettaglio partita
`/play-records/[id]/edit`	Modifica partita
`/play-records/stats`	Statistiche partite

Giocatori

URL	Descrizione
`/players`	Lista giocatori
`/players/[id]`	Profilo giocatore
`/players/[id]/achievements`	Achievements giocatore *(nuovo)*
`/players/[id]/games`	Giochi del giocatore *(nuovo)*
`/players/[id]/sessions`	Sessioni del giocatore *(nuovo)*
`/players/[id]/stats`	Statistiche giocatore *(nuovo)*

Profilo & Impostazioni

URL	Descrizione
`/profile`	Profilo utente con tabs *(nuovo)*
`/profile/achievements`	Achievements utente
`/settings/notifications`	Notifiche
`/settings/security`	Sicurezza e 2FA
`/setup`	Setup iniziale

Toolkit

URL	Descrizione
`/toolkit`	Toolkit principale
`/toolkit/demo`	Demo toolkit
`/toolkit/history`	Storico toolkit
`/toolkit/[sessionId]`	Sessione toolkit attiva

Knowledge Base & Notifiche

URL	Descrizione
`/knowledge-base/[id]`	Voce KB specifica
`/notifications`	Centro notifiche
`/badges`	Badge e achievement system
`/upload`	Upload file/documenti
`/n8n`	n8n workflow integration

Editor (ruolo Editor)

URL	Descrizione
`/editor`	Dashboard editor
`/editor/agent-proposals`	Proposte agenti
`/editor/agent-proposals/create`	Crea proposta agente
`/editor/agent-proposals/[id]/edit`	Modifica proposta
`/editor/agent-proposals/[id]/test`	Test proposta

Admin — Nuovo Dashboard Unificato

Overview

URL	Descrizione
`/admin/overview`	Dashboard principale admin
`/admin/overview/activity`	Activity log
`/admin/overview/system`	Stato sistema

Agenti AI

URL	Descrizione
`/admin/agents`	Hub gestione agenti
`/admin/agents/builder`	Creazione/modifica agenti *(nuovo)*
`/admin/agents/analytics`	Analytics agenti
`/admin/agents/models`	Configurazione modelli AI
`/admin/agents/pipeline`	Visualizzazione pipeline RAG *(nuovo #4624)*
`/admin/agents/debug`	Debug pipeline RAG *(nuovo #4624)*
`/admin/agents/debug-chat`	Chat con debug RAG *(nuovo)*
`/admin/agents/strategy`	Configurazione strategie RAG *(nuovo #4624)*
`/admin/agents/chat-history`	Storico chat agenti *(nuovo)*
`/admin/agents/chat-limits`	Limiti sessioni chat *(nuovo #4918)*

Knowledge Base

URL	Descrizione
`/admin/knowledge-base`	Hub gestione KB
`/admin/knowledge-base/documents`	Gestione PDF indicizzati *(nuovo #4787)*
`/admin/knowledge-base/vectors`	Collezioni vettoriali Qdrant *(nuovo #4786)*
`/admin/knowledge-base/pipeline`	Pipeline elaborazione
`/admin/knowledge-base/queue`	Coda documenti *(nuovo #4892)*
`/admin/knowledge-base/processing`	Stato elaborazione attuale
`/admin/knowledge-base/embedding`	Configurazione embedding
`/admin/knowledge-base/upload`	Upload PDF admin *(nuovo #4925)*
`/admin/knowledge-base/settings`	Impostazioni KB *(nuovo #4888)*

Shared Games

URL	Descrizione
`/admin/shared-games`	Hub gestione shared games
`/admin/shared-games/new`	Crea shared game
`/admin/shared-games/all`	Tutti i shared games
`/admin/shared-games/categories`	Gestione categorie
`/admin/shared-games/[id]`	Dettaglio shared game

Utenti

URL	Descrizione
`/admin/users`	Dashboard gestione utenti
`/admin/users/activity`	Attività utenti
`/admin/users/roles`	Gestione ruoli
`/admin/users/[id]`	Dettaglio utente

Admin — Struttura Legacy (ancora attiva)

URL	Descrizione
`/admin/agent-definitions`	Definizioni agenti
`/admin/agent-definitions/create`	Crea definizione agente
`/admin/agent-definitions/[id]/edit`	Modifica definizione
`/admin/agent-definitions/playground`	Playground agenti
`/admin/agent-typologies`	Tipologie agenti
`/admin/agents/catalog`	Catalogo agenti
`/admin/agents/metrics`	Metriche agenti
`/admin/ai-lab/agents/builder`	Builder avanzato AI Lab
`/admin/ai-lab/multi-agent`	Sistema multi-agente
`/admin/ai-models`	Modelli AI
`/admin/ai-usage`	Utilizzo API AI
`/admin/alerts`	Alert attivi
`/admin/alert-rules`	Regole alert
`/admin/analytics`	Analytics sistema
`/admin/rag-executions`	Log esecuzioni RAG
`/admin/rag/execution-replay`	Replay operazioni RAG
`/admin/rag/strategy-builder`	Builder strategie RAG
`/admin/rag/tier-strategy-config`	Config strategie per tier
`/admin/games`	Gestione database giochi
`/admin/games/import/bulk`	Import bulk giochi
`/admin/games/import/wizard`	Wizard import giochi
`/admin/game-sessions`	Gestione sessioni
`/admin/pdfs`	Gestione PDF
`/admin/faqs`	Gestione FAQ
`/admin/sessions`	Sessioni admin

`/admin/configuration`	Configurazione sistema
`/admin/configuration/game-library-limits`	Limiti libreria giochi
`/admin/configuration/pdf-tier-limits`	Limiti PDF per tier
`/admin/feature-flags`	Feature flags
`/admin/tier-limits`	Limitazioni per tier
`/admin/prompts`	Gestione prompt di sistema
`/admin/strategies`	Strategie
`/admin/audit-log`	Audit trail azioni
`/admin/infrastructure`	Monitoraggio infrastruttura
`/admin/cache`	Gestione cache
`/admin/services`	Stato microservizi
`/admin/api-keys`	Gestione API key
`/admin/reports`	Report
`/admin/bulk-export`	Export dati
`/admin/share-requests`	Richieste di condivisione
`/admin/n8n-templates`	Template n8n

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3. Endpoint Backend Aggiunti

Game Management

Metodo	Path	Descrizione	Issue
`GET`	`/api/v1/games/{id}/reviews`	Recensioni del gioco	#4904
`GET`	`/api/v1/games/{id}/strategies`	Strategie per il gioco	#4903
`GET`	`/api/v1/games/{id}/preview`	Preview gioco per wizard	#4776

Knowledge Base / RAG

Metodo	Path	Descrizione	Issue
`GET`	`/api/v1/knowledge-base/{gameId}/status`	Stato indicizzazione PDF per game	#4943
`GET`	`/api/v1/knowledge-base/agent-slots`	Slot agenti disponibili (quota tier)	#4799
`POST`	`/api/v1/knowledge-base/agents`	Orchestrated agent creation con auto-setup	#4800

Admin — PDF & Documents

Metodo	Path	Descrizione	Issue
`POST`	`/api/v1/admin/shared-games/{id}/pdfs`	Upload PDF per shared game (500MB, bypass quota)	#4922
`GET`	`/api/v1/admin/shared-games/{id}/agent`	Agente collegato allo shared game	#4923
`GET`	`/api/v1/admin/shared-games/{id}/kb-cards`	KB cards dello shared game	#4924

Admin — Agent Definitions

Metodo	Path	Descrizione	Issue
`PUT`	`/api/v1/admin/agent-definitions/{id}/kb-card-ids`	Aggiorna KB cards dell'agente (JSONB)	#4925

Live Game Session

Metodo	Path	Descrizione	Issue
`POST`	`/api/v1/game-sessions`	Crea sessione live	#4749
`GET`	`/api/v1/game-sessions/{id}`	Dettaglio sessione	#4749
`PUT`	`/api/v1/game-sessions/{id}/join`	Unisciti sessione via codice	#4766
`POST`	`/api/v1/game-sessions/{id}/actions`	Azione giocatore	#4765
`GET`	`/api/v1/game-sessions/{id}/state`	Stato corrente sessione	#4749
`GET`	`/api/v1/game-sessions/{id}/snapshots`	Storico snapshot (time travel)	#4755
`GET`	`/api/v1/game-sessions/active`	Sessioni attive	#4749
`GET`	`/api/v1/game-sessions/{id}/sse`	SSE stream stato sessione	#4764

Notifications

Metodo	Path	Descrizione	Issue
`POST`	`/api/v1/notifications/processing-jobs`	Notifica multi-channel per eventi elaborazione	#4744

Chat / Agent Proxy

Metodo	Path	Descrizione	Issue
`POST`	`/api/chat/stream`	Proxy SSE OpenRouter per streaming chat	#4806

Riepilogo Numerico

Area	Nuovi/Modificati
Pagine frontend nuove	~25
Pagine admin nuove	~15
Endpoint backend nuovi	~20
Componenti UI (MeepleCard system)	~10
Epiche completate	4

Epiche: #4789 (RAG Admin), #4820 (Admin SharedGame), #4912 (Custom Agents/Chat), #4757 (Player Journey parziale)

features/pdf-processing-queue-dashboard.md

PDF Processing Queue Dashboard - Design Brief

Date: 2026-02-19 **Status**: Approved (Brainstorm session) **Scope**: Backend queue infrastructure (Quartz) + Admin dashboard UI

Overview

Dashboard amministrativa per monitorare e gestire la pipeline di elaborazione PDF → embedding. Sostituisce la pagina mock esistente a `/admin/knowledge-base` con una dashboard funzionale che mostra lo stato della coda in tempo reale.

Decisioni Architetturali

Aspetto	Decisione	Rationale
Target utenti	Admin/Editor (utente finale riceve notifica)	Separazione ruoli
Queue engine	Quartz v1 (già presente), Temporal v2 futuro	Pragmatismo, zero nuove dipendenze
UI pattern	Lista compatta + detail panel on click	Efficienza informativa
Route	`/admin/knowledge-base` (sostituisce mock esistente)	Evoluzione naturale
Azioni coda	Add, Remove, Retry, Priorita (drag&drop), Cancel	Controllo completo
Real-time	SSE (endpoint gia esistenti)	Infrastruttura pronta
Notifiche	In-app + Email + Push a fine processing	Multi-canale
Persistenza	PostgreSQL (Quartz AdoJobStore) - sopravvive al restart	Affidabilita
Limiti	Max 100 PDF in coda, 50MB per file, 3 job paralleli	Capacita controllata
Log retention	Sempre, job completati visibili ma filtrabili	Storico completo
Re-processing	Solo full-retry, no step parziale (v1)	Semplicita

Architettura

(blocco di codice rimosso)

Backend Design

Domain: ProcessingJob Aggregate

(blocco di codice rimosso)

Domain Events

(blocco di codice rimosso)

Application Layer

(blocco di codice rimosso)

Quartz Configuration

(blocco di codice rimosso)

API Endpoints

(blocco di codice rimosso)

File Structure (Backend)

(blocco di codice rimosso)

Frontend Design

Layout

(blocco di codice rimosso)

Filters

(blocco di codice rimosso)

Component Structure

(blocco di codice rimosso)

Dependencies

(blocco di codice rimosso)

Notifications

Trigger Flow

(blocco di codice rimosso)

Recipients

- **Completed**: User who uploaded the PDF
- **Failed**: User who uploaded + all Admins (for visibility)

Implementation Plan

Phase	Scope	Labels
1	ProcessingJob aggregate + EF migration + Quartz config	backend, area/pdf
2	Commands/Queries/Handlers + API endpoints	backend, area/pdf, area/api
3	SSE streaming for queue + single job	backend, area/pdf
4	Frontend: queue list + detail panel + filters	frontend, area/admin, area/ui
5	Frontend: drag & drop + actions (cancel/retry/remove)	frontend, area/admin, area/ui
6	Frontend: SSE integration + real-time updates	frontend, area/admin, area/ui
7	Notifications: in-app + email + push	backend, frontend
8	Tests + code review	tests, area/pdf

Constraints

- Max 100 PDF in queue simultaneously
- Max file size: 50MB per PDF
- Max 3 parallel processing jobs
- Jobs persist across server restarts (Quartz PostgreSQL store)
- Log retention: forever, filterable by status/date
- v1: Full retry only (no partial step re-processing)
- v2 future: Migrate to Temporal for durable workflows

Existing Infrastructure to Leverage

- **Quartz 3.13.1**: Already in Api.csproj
- **SSE endpoints**: `/pdfs/{id}/progress/stream`, `/pdfs/{id}/status/stream`
- **Admin UI shell**: AdminShell, AdminSidebar, AdminTopNav
- **Backend API**: PDF upload (chunked), progress tracking, admin CRUD

- **Domain events**: Event-driven architecture already in place
- **UserNotifications BC**: Exists (needs verification of implementation level)

References

- ADR-024: RAG Pipeline Enhancement Roadmap
- Existing: `EnhancedPdfProcessingOrchestrator.cs`
- Existing: `ExtractPdfTextCommandHandler.cs`
- Existing: `/admin/knowledge-base/upload` (mock UI to replace)